MySchola

GCSE BIOLOGY

# TOPIC 2: ORGANISATION

Detailed AQA Specification Notes

**Clear explanations • Required practicals • Maths skills • Exam tips**

**HOW TO USE THIS BOOKLET**

Learn each section, then use the exam tips and check yourself questions to test whether you can explain the biology in full sentences. The content follows the uploaded AQA Topic 2 specification and expands each required point to an exam ready GCSE level.

*MySchola • Learn clearly. Practise purposefully.*

# Topic map

*Specification focus: Everything in AQA Topic 2: Organisation at a glance*

| Section | Focus | What you must be able to do |
|---|---|---|
| 1 | Principles of organisation | Cells, tissues, organs, organ systems and organisms |
| 2 | Digestive system and enzymes | Enzyme action, digestive enzymes, bile and absorption |
| 3 | Required Practical 4 | Food tests for sugars, starch and protein, plus lipid testing |
| 4 | Required Practical 5 | Effect of pH on the rate of amylase activity |
| 5 | Heart, lungs and circulation | Double circulation, heart structures and gas exchange |
| 6 | Blood vessels and blood | Arteries, veins, capillaries and blood components |
| 7 | Cardiovascular disease | CHD, stents, statins, valves, transplants and artificial hearts |
| 8 | Health and disease | Interactions between disease and interpreting epidemiological data |
| 9 | Risk factors and lifestyle | Smoking, alcohol, diet, obesity, exercise and carcinogens |
| 10 | Cancer | Benign and malignant tumours, spread and risk factors |
| 11 | Plant tissues and the leaf | Epidermal, mesophyll, xylem, phloem, meristems and stomata |
| 12 | Plant transport | Root hair cells, transpiration, translocation and environmental factors |
| 13 | Maths and exam checklist | Rates, graphs, sampling, correlation and revision checklist |

**SPECIFICATION FLAGS**

REQUIRED PRACTICAL means you should know the method, variables, safety and how to interpret results. The uploaded extract does not explicitly label any content here as HT ONLY, so no Higher Tier only flag has been added.

# 1 Principles of organisation

*Specification focus: cells, tissues, organs, organ systems and size/scale*

Living organisms are organised in levels. The key idea is that specialised cells work together, and each higher level combines structures to carry out larger functions.

| Level | Definition | Example |
|---|---|---|
| Cell | The basic building block of a living organism. | A muscle cell |
| Tissue | A group of cells with a similar structure and function. | Muscle tissue |
| Organ | An aggregation of different tissues performing specific functions. | The heart |
| Organ system | A group of organs working together to perform a major function. | The digestive system |
| Organism | The complete living individual formed from organ systems. | A human |

**EXAM LANGUAGE**

Do not describe an organ as “a group of cells”. An organ contains several tissues. A tissue is the level that is defined as a group of cells with a similar structure and function.

## Applying size and scale

Questions can ask you to compare the relative sizes of cells, tissues, organs and organ systems. Read the units carefully and remember that the organisational level becomes larger as you move from cell to tissue to organ to organ system.

**CHECK YOURSELF**

Put these in order from smallest to largest: organ, cell, organ system, tissue. Answer: cell → tissue → organ → organ system.

# 2 Digestive system and enzymes

*Specification focus: enzyme molecules, digestive enzymes, bile and products of digestion*

The digestive system is an organ system in which several organs work together to digest food and absorb the small soluble products into the bloodstream.

## What enzymes do

- Enzymes are biological catalysts. They increase the rate of specific chemical reactions in living organisms.
- Each enzyme has an active site with a particular shape. A substrate with a complementary shape fits into the active site.
- The lock and key model is a simplified way to explain why enzymes are specific to particular substrates.
- Digestive enzymes break large, insoluble food molecules into smaller, soluble molecules that can be absorbed into the blood.

**KEY IDEA**

The active site is central to exam answers. When temperature or pH changes too far from the optimum, the active site can change shape so the substrate no longer fits effectively.

## Digestive enzymes you must know

| Enzyme group | Acts on | Products | Key specification point |
|---|---|---|---|
| Carbohydrases | Carbohydrates | Simple sugars | Amylase is a carbohydrase that breaks down starch. |
| Proteases | Proteins | Amino acids | Protein digestion produces amino acids that can be absorbed. |
| Lipases | Lipids (fats) | Glycerol and fatty acids | Lipase acts more quickly when fat has been emulsified into small droplets. |

### Sites of production and action you must recall

| Enzyme | Produced by | Main site(s) of action |
|---|---|---|
| Amylase | Salivary glands, pancreas and small intestine | Mouth and small intestine |
| Proteases | Stomach, pancreas and small intestine | Stomach and small intestine |
| Lipases | Pancreas and small intestine | Small intestine |

### Simple word equations

**starch → simple sugars | protein → amino acids | lipid → glycerol + fatty acids**

In an exam, you may include the enzyme name over the arrow, for example: starch —amylase→ simple sugars.

## What happens to the products?

- The products of digestion can be used to build new carbohydrates, lipids and proteins in the body.
- Some glucose is used in respiration to release energy for cellular processes.

# 2A Enzyme activity and bile

*Specification focus: temperature, pH, rate and the role of bile*

## Temperature and enzyme activity

At low temperatures particles have less kinetic energy, so successful collisions between enzyme and substrate happen less often. As temperature rises, reaction rate increases until the enzyme reaches an optimum. Above the optimum, the enzyme can denature: its active site changes shape and the substrate no longer fits properly.

## pH and enzyme activity

Each enzyme works best at an optimum pH. Moving too far away from the optimum changes the conditions around the enzyme and can alter the active site. This lowers the rate because fewer enzyme substrate complexes form.

**EXAM PRECISION**

Do not say the enzyme “dies”. Enzymes are molecules, not cells. Use the phrase “the enzyme denatures and the active site changes shape”.

## Bile

Bile is made in the liver and stored in the gall bladder. It helps lipid digestion in two ways:

| Action of bile | Why it matters |
|---|---|
| It is alkaline. | It neutralises hydrochloric acid from the stomach, producing conditions that allow digestive enzymes in the small intestine to work more effectively. |
| It emulsifies fat into small droplets. | This increases the surface area of fat available to lipase, increasing the rate of fat breakdown. |

**RATE CALCULATION**

Rate = amount of change ÷ time. If a reaction is measured by the time taken to reach an endpoint, a useful comparison is rate = 1 ÷ time. A shorter time therefore means a faster rate.

# 3 Required Practical 4: food tests

*Specification focus: qualitative tests for carbohydrates, lipids and proteins*

**REQUIRED PRACTICAL**

Use qualitative reagents to test for a range of carbohydrates, lipids and proteins. The specification explicitly names Benedict's test for sugars, iodine for starch and Biuret reagent for protein.

## Tests and positive results

| Food substance | Reagent / method | Positive result |
|---|---|---|
| Reducing sugars | Add Benedict's reagent and heat the sample in a hot water bath. | Blue can change through green/yellow/orange to brick red depending on the amount of reducing sugar. |
| Starch | Add iodine solution to the sample. | Orange/brown changes to blue black. |
| Protein | Add Biuret reagent to the sample. | Blue changes to lilac/purple. |
| Lipids | Add Sudan III stain to the prepared food solution and shake gently. | A red stained oil layer separates and floats at the surface if lipid is present. |

*The uploaded specification names Benedict's, iodine and Biuret. It does not name the lipid reagent; AQA's required practical guidance uses Sudan III stain.*

## A good practical method

1. Place a small, equal volume of each food sample into labelled test tubes.
2. Use clean apparatus so samples are not contaminated by substances from previous tests.
3. Add the correct reagent for the substance being tested.
4. For Benedict's test, heat the tube safely in a water bath rather than directly in a flame.
5. Record the final colour and compare it with the expected positive and negative result.
6. Repeat where appropriate to improve confidence in the observation.

**SAFETY**

Wear eye protection. Benedict's test involves hot water and hot glassware. Sudan III contains ethanol and is flammable, so keep it away from naked flames.

**EXAM TIP**

Qualitative means the test tells you whether a substance is present from an observed result such as a colour change. Do not automatically describe these tests as giving an exact concentration.

# 4 Required Practical 5: pH and amylase

*Specification focus: continuous sampling, iodine testing and controlling temperature*

This practical investigates how pH affects the rate at which amylase digests starch. Iodine is used to test whether starch is still present.

## Method

7. Place separate drops of iodine solution into the wells of a spotting tile.
8. Use a water bath or electric heater to keep the reaction mixture at a constant temperature.
9. Add amylase and a buffer solution of a chosen pH to a test tube and allow them to reach the controlled temperature.
10. Add starch solution, mix and start the timer immediately.
11. Every 30 seconds, remove a small sample of the reaction mixture and add it to a fresh iodine drop.
12. Record the time when the iodine no longer turns blue black. At this point the starch has been completely digested.
13. Repeat for a range of pH values while keeping the other important variables constant.

## Variables

| Type | Variable |
|---|---|
| Independent | pH of the reaction mixture. |
| Dependent | Time taken for starch to disappear, or rate calculated from 1 ÷ time. |
| Control | Temperature, concentration and volume of amylase, concentration and volume of starch, total reaction volume and sampling interval. |

**WHY THE WATER BATH MATTERS**

Temperature affects enzyme activity. If temperature changed at the same time as pH, you could not tell whether a change in reaction rate was caused by pH or temperature.

## Worked rate example

If the starch disappears after 120 s: rate = 1 ÷ 120 = 0.0083 $s^{-1}$. A reaction that finishes in 60 s has a higher rate because 1 ÷ 60 is larger.

**COMMON ERROR**

The endpoint is not “when iodine becomes blue black”. Blue black means starch is still present. The endpoint is the first sample where iodine remains its original orange/brown colour.

# 5 Heart, lungs and double circulation

*Specification focus: heart structure, major blood vessels, pacemakers and gas exchange*

The heart is an organ that pumps blood around the body in a double circulatory system. Blood passes through the heart twice during one complete circuit of the body.

| Side / chamber | Main job |
|---|---|
| Right ventricle | Pumps blood to the lungs, where gas exchange takes place. |
| Left ventricle | Pumps blood around the rest of the body. It works against a greater resistance, so it has a thicker muscular wall. |

## Heart structure and one way flow

- The atria receive blood into the heart and the ventricles pump blood out of the heart.
- The left ventricle has a thicker muscular wall than the right ventricle because it must generate enough pressure to pump blood around the whole body.
- Valves help keep blood moving in one direction by preventing backflow. You do not need to learn the individual names of the heart valves for this specification.

## Blood vessels associated with the heart

| Vessel | Direction / function |
|---|---|
| Aorta | Carries oxygenated blood away from the left ventricle to the body. |
| Vena cava | Returns deoxygenated blood from the body to the heart. |
| Pulmonary artery | Carries deoxygenated blood from the heart to the lungs. |
| Pulmonary vein | Carries oxygenated blood from the lungs back to the heart. |
| Coronary arteries | Supply the heart muscle itself with oxygenated blood. |

**ROUTE TO LEARN**

Body → vena cava → right side of heart → pulmonary artery → lungs → pulmonary vein → left side of heart → aorta → body.

## Lungs and gas exchange

- The specification limits lung structures to the trachea, bronchi, alveoli and the capillary network surrounding the alveoli.
- Alveoli provide a large surface area for gas exchange and have thin walls, giving a short diffusion distance.
- A rich capillary supply maintains concentration gradients for oxygen and carbon dioxide.

## Heart rate control

The natural resting heart rate is controlled by a group of cells in the right atrium that act as a pacemaker. An artificial pacemaker is an electrical device used to correct irregularities in heart rate.

# 6 Blood vessels and blood

*Specification focus: structure linked to function and the four blood components*

## Blood vessels

| Vessel | Structure | How structure supports function |
|---|---|---|
| Artery | Thick muscular and elastic walls; relatively small lumen. | Carries blood away from the heart at high pressure. Strong walls withstand pressure and elastic tissue helps maintain flow. |
| Vein | Thinner walls; larger lumen; valves. | Returns blood to the heart at lower pressure. Valves help prevent backflow. |
| Capillary | Wall is one cell thick; very narrow lumen; forms networks in tissues. | Short diffusion distance allows rapid exchange of substances between blood and cells. |

**DO NOT MEMORISE DIRECTION INCORRECTLY**

Arteries are defined by carrying blood away from the heart and veins by carrying blood towards the heart. The pulmonary artery and pulmonary vein are important reminders that oxygenation is not the definition.

## Blood is a tissue

Blood consists of plasma with red blood cells, white blood cells and platelets suspended in it.

| Component | Function / adaptation |
|---|---|
| Plasma | Transports blood cells and dissolved substances around the body. |
| Red blood cells | Transport oxygen. Their shape gives a large surface area and they contain haemoglobin for oxygen transport. |
| White blood cells | Defend the body against pathogens as part of the immune response. |
| Platelets | Cell fragments involved in blood clotting, helping to reduce blood loss and entry of microorganisms after injury. |

**RECOGNITION SKILL**

You may be shown a photograph or diagram of blood cells and asked to identify them or explain how a feature is adapted to function. Always link the visible feature to what the cell must do.

# 7 Cardiovascular disease and treatments

*Specification focus: CHD, stents, statins, faulty valves, transplants and artificial hearts*

## Coronary heart disease (CHD)

In CHD, layers of fatty material build up inside the coronary arteries. This narrows the arteries and reduces blood flow to the heart muscle, so the muscle receives less oxygen for respiration.

| Treatment | How it helps | Advantages | Disadvantages / risks |
|---|---|---|---|
| Stent | Keeps a narrowed coronary artery open so blood can flow more easily. | Rapidly improves blood flow; can reduce the risk of a heart attack. | Procedure has risks and the artery can narrow again. |
| Statins | Reduce blood cholesterol, slowing the rate of fatty material deposition. | Can lower long term cardiovascular risk. | Must usually be taken regularly and can cause side effects. |
| Valve replacement | Replaces a valve that does not open fully or leaks. | Restores more effective one way blood flow. | Surgery carries risk; mechanical and biological valves each have limitations. |
| Heart transplant | A donor heart replaces a failing heart. | Can restore heart function when other treatments are unsuitable. | Donor shortage, major surgery and risk of rejection. |
| Artificial heart | Temporarily supports circulation while waiting for a transplant or allows the heart to rest. | Can keep a patient alive while another solution is found. | Risk of complications such as clotting or infection; not a simple permanent cure. |

**EVALUATION QUESTIONS**

For a 4 to 6 mark treatment question, compare benefits and risks rather than simply listing features. Use the patient context if one is given.

## Faulty valves

A faulty heart valve may fail to open fully, restricting blood flow, or may leak so some blood moves backwards. Biological or mechanical replacement valves can be used.

# 8 Health, disease and interactions

*Specification focus: physical and mental wellbeing, interactions and epidemiological data*

Health is the state of physical and mental wellbeing. Ill health can be caused by communicable disease, non communicable disease and other factors such as diet, stress and life situations.

## Diseases can interact

| Interaction | Explanation |
|---|---|
| Immune system defects and infection | If the immune system does not function effectively, the person is more likely to suffer from infectious diseases. |
| Viruses and cancer | Viruses living in cells can trigger changes that contribute to cancers. |
| Pathogens and allergies | Immune reactions initially triggered by a pathogen can lead to allergies such as skin rashes or asthma. |
| Physical illness and mental illness | Severe physical ill health can contribute to depression and other mental illness. |

## Interpreting health data

- Translate disease incidence information between numerical and graphical forms.
- Construct and interpret frequency tables, bar charts and histograms.
- Use scatter diagrams to identify whether two variables show a correlation.
- Understand that sampling must be representative if conclusions are to apply to a wider population.

**CORRELATION IS NOT CAUSATION**

A scatter graph can show that two variables are associated, but this alone does not prove that one variable causes the other. Other variables may be involved.

# 9 Risk factors and lifestyle

*Specification focus: links to non communicable disease and evaluating evidence*

A risk factor is linked to an increased rate of a disease. It may be an aspect of lifestyle or a substance in a person's body or environment. For some risk factors a causal mechanism is known, while for others only an association may be established.

| Risk factor | Specified link |
|---|---|
| Diet, smoking and exercise | Cardiovascular disease. |
| Obesity | Increased risk of Type 2 diabetes. |
| Alcohol | Effects on the liver and brain function. |
| Smoking | Lung disease and lung cancer. |
| Smoking and alcohol in pregnancy | Can harm unborn babies. |
| Carcinogens, including ionising radiation | Increase the risk of cancer. |

## Why many diseases are difficult to explain

Many diseases result from the interaction of several factors. A person may have more than one lifestyle risk factor and may also have genetic or environmental influences. This is why carefully designed studies and appropriate sampling are important.

## Human and financial cost

- For the individual: reduced quality of life, disability, treatment burden and loss of income.
- For communities and nations: healthcare costs and loss of productivity when people are unable to work.
- At global level: large differences in disease burden can reflect different exposure to risk factors and access to prevention or treatment.

**DATA QUESTION STRATEGY**

Identify the pattern, support it with figures from the data, then state whether the evidence shows correlation or supports a causal explanation. Do not claim more than the data shows.

# 10 Cancer

*Specification focus: uncontrolled cell growth, benign and malignant tumours*

Cancer results from changes in cells that lead to uncontrolled growth and division.

| Tumour type | What happens |
|---|---|
| Benign | A growth of abnormal cells contained in one area, usually within a membrane. It does not invade other parts of the body. |
| Malignant | Cancer cells invade neighbouring tissues and can spread to different parts of the body in the blood, where they may form secondary tumours. |

## Risk factors for cancer

- Scientists have identified lifestyle risk factors for a range of cancers.
- There are also genetic risk factors for some cancers.
- The wider Topic 2 risk factor section identifies carcinogens, including ionising radiation, as cancer risk factors and smoking as a risk factor for lung cancer.

**EXAM LANGUAGE**

“Benign” does not mean the cells are normal. They are abnormal cells, but they remain contained. “Malignant” means the tumour can invade surrounding tissue and spread to other parts of the body.

## Explaining secondary tumours

A strong answer explains the sequence: malignant cells invade nearby tissue → cells can enter the blood → they travel to another part of the body → they divide there and form a secondary tumour.

# 11 Plant tissues and the leaf

*Specification focus: specialised tissues and how structure relates to function*

The leaf is a plant organ made from several tissues. The specification limits the leaf structures you need to know to epidermis, palisade and spongy mesophyll, xylem, phloem and guard cells surrounding stomata.

| Plant tissue / structure | Function |
|---|---|
| Epidermal tissue | Covers the plant surface and forms a protective outer layer. |
| Palisade mesophyll | Main photosynthetic tissue in the leaf; cells are positioned to receive light efficiently. |
| Spongy mesophyll | Contains spaces that allow gases to move through the leaf for gas exchange. |
| Xylem | Transports water and mineral ions from roots towards stems and leaves. |
| Phloem | Transports dissolved sugars around the plant by translocation. |
| Meristem tissue | Found at growing tips of roots and shoots; contains cells capable of division and differentiation. |
| Guard cells and stomata | Guard cells control the opening of stomata, regulating gas exchange and water loss. |

**OBSERVATION SKILL**

You may be asked to recognise or draw a transverse section of a leaf. Label only structures you can justify from the diagram and link their position or structure to their function.

## The plant transport organ system

Roots, stems and leaves work together as a plant organ system for transporting substances. Root tissues absorb water and mineral ions, the stem contains transport tissues, and leaves exchange gases and lose water vapour through stomata.

# 12 Plant transport

*Specification focus: root hair cells, xylem, phloem, transpiration and translocation*

## Root hair cells

Root hair cells are adapted for efficient uptake. Their long extension provides a large surface area. Water enters by osmosis, while mineral ions can be absorbed by active transport when their concentration is lower in the soil than inside the root cells.

## Xylem

Xylem tissue transports water and mineral ions from the roots to the stems and leaves. It forms hollow tubes strengthened by lignin, making the tissue well suited for carrying water in the transpiration stream.

## Phloem

Phloem tissue transports dissolved sugars from the leaves to the rest of the plant for immediate use or storage. This movement of food molecules is called translocation. The specification requires phloem to be recognised as tubes of elongated cells with pores in the end walls, but detailed transport mechanisms are not required.

| Process | Substance moved | Main route / direction |
|---|---|---|
| Transpiration | Water | Water moves from roots through xylem and is lost as water vapour from leaves. |
| Translocation | Dissolved sugars | Sugars move through phloem from leaves to parts of the plant where they are used or stored. |

**DO NOT MIX THEM UP**

Xylem = water and mineral ions. Phloem = dissolved sugars. Transpiration describes water loss and movement through the plant; translocation describes movement of dissolved food molecules through phloem.

# 12A Transpiration and stomata

*Specification focus: environmental factors, guard cells and rate calculations*

Stomata are pores in the leaf surface. Guard cells control their opening and closing, helping to balance carbon dioxide uptake for photosynthesis with water loss by transpiration.

| Factor | Typical effect on transpiration rate | Reason |
|---|---|---|
| Higher temperature | Increases rate. | Water molecules have more kinetic energy and evaporation from leaf surfaces is faster. |
| Lower humidity | Increases rate. | A larger difference in water vapour concentration exists between the leaf air spaces and the outside air. |
| Greater air movement | Increases rate. | Moving air removes water vapour from around the leaf, maintaining a steep concentration gradient. |
| Greater light intensity | Usually increases rate. | Stomata tend to open more to allow carbon dioxide in for photosynthesis, increasing opportunities for water loss. |

## Measuring transpiration

The specification expects you to measure transpiration indirectly by water uptake. A potometer can be used to record how quickly water is taken up by a cut shoot. Water uptake is used as an estimate of transpiration rate.

**RATE EQUATION**

Rate of water uptake = volume of water taken up ÷ time. If an air bubble moves along a capillary tube, distance moved per unit time can also be used as a comparative measure.

## Stomata investigations

- Investigations can compare the distribution of stomata and guard cells on different leaf surfaces.
- Use appropriate sampling rather than counting only one tiny area if you want a representative estimate.
- Calculate an arithmetic mean when several counts or rate measurements are available.
- Plot graphs with appropriate scales and units, then describe the pattern before explaining it.

# 13 Maths and data skills

*Specification focus: rate, means, sampling, graphs and correlation*

## Core calculations

| Skill | Equation / method | Example |
|---|---|---|
| Reaction rate | Rate = amount of change ÷ time | 30 units of change in 60 s gives 0.5 units $s^{-1}$. |
| Enzyme endpoint rate | Rate = 1 ÷ time | Endpoint at 100 s gives 0.010 $s^{-1}$. |
| Blood flow rate | Rate = volume ÷ time | 300 $cm^3$ in 60 s gives 5 $cm^3$ $s^{-1}$. |
| Transpiration / water uptake | Rate = water uptake ÷ time | 2.4 $cm^3$ in 30 min gives 0.08 $cm^3$ $min^{-1}$. |
| Arithmetic mean | Add values ÷ number of values | (12 + 15 + 18) ÷ 3 = 15. |
| Surface area / volume | Use the shape formula given or known, then keep units consistent. | Compare values fairly using the same units. |

## Graph and table skills

- Choose a sensible scale that uses most of the graph area.
- Put the independent variable on the x axis and dependent variable on the y axis unless told otherwise.
- Include quantity and unit in each axis label.
- For continuous variables, use an appropriate line graph or scatter graph rather than separate category bars.
- Use a scatter diagram to identify positive correlation, negative correlation or no clear correlation.
- When interpreting a graph, quote data values to support the pattern you describe.

**SAMPLING**

A larger, representative sample generally makes a conclusion more reliable because it is less likely to be dominated by unusual individuals or a single unrepresentative area.

# Final revision checklist

*Specification focus: what you should be able to explain without notes*

| Checklist A | Checklist B |
|---|---|
| ☐ Define tissue, organ and organ system and place levels of organisation in order. | ☐ Explain enzyme specificity using the active site and lock and key model. |
| ☐ Explain how temperature and pH affect enzyme activity. | ☐ Recall what amylase, proteases and lipases break down and the products formed. |
| ☐ Explain both roles of bile: neutralising acid and emulsifying fat. | ☐ Describe the tests for sugars, starch, protein and lipids with positive results. |
| ☐ Describe Required Practical 5 and identify independent, dependent and control variables. | ☐ Describe the route of blood through the double circulatory system and name the five specified vessels. |
| ☐ Compare arteries, veins and capillaries by linking structure to function. | ☐ State the functions of plasma, red blood cells, white blood cells and platelets. |
| ☐ Explain CHD and evaluate stents, statins, replacement valves, transplants and artificial hearts. | ☐ Define health and explain the specified interactions between diseases. |
| ☐ Explain what a risk factor is and distinguish correlation from causation. | ☐ Describe benign and malignant tumours and how secondary tumours form. |
| ☐ Explain the functions of epidermis, mesophyll, xylem, phloem, meristems and guard cells. | ☐ Explain root hair cell adaptations, transpiration and translocation. |
| ☐ Predict how temperature, humidity, air movement and light intensity affect transpiration. | ☐ Calculate rates and means and interpret tables, bar charts, histograms and scatter graphs. |

## Exam style quick questions

1. Explain why bile increases the rate of lipid digestion. [3 marks]
2. Describe how you would investigate the effect of pH on amylase activity. [6 marks]
3. Explain how the structure of an artery is related to its function. [3 marks]
4. A graph shows smoking frequency and lung disease incidence rising together. What can you conclude and what can you not conclude from the graph alone? [3 marks]
5. Explain how xylem and phloem differ in the substances they transport. [2 marks]
6. Explain why increasing air movement usually increases the rate of transpiration. [2 marks]

**LAST CHECK**

For longer answers, use precise biological terms, link structure to function and explain cause and effect. When data is provided, use figures from the table or graph rather than making a vague statement.